

DPL Pvt Ltd – FBR Digital Invoicing Integration Plan

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System: QuickBooks / ERP / Finance Team Tool

Integration Target: FBR Digital Invoicing API

1. Objective

The goal of this integration is to enable DPL's Finance team to log invoices directly into FBR in a controlled, secure, and auditable manner. Key objectives include:

- **Real-time invoice recording** in FBR
- **Ensuring data validation** and compliance
- **Minimizing errors** and simplifying Finance workflow

2. Current Understanding & Approach Options

Based on the FBR Digital Invoicing documentation, the following approaches are possible:

Approach	Description	Pros	Cons
Finance Team Tool	Desktop app or Streamlit web app allowing team to enter invoices	User-friendly; automatic FBR response	Requires development
Postman / API Client	Manual JSON payload submission to FBR	Fast to implement	Less user-friendly; manual effort
AWS Integration	Lambda + API Gateway + DynamoDB + S3 + CloudWatch	Fully automated, scalable, monitored	Higher complexity; requires AWS setup

Next Step: Identify exact requirements before choosing the approach.

3. Integration Requirements (High-Level)

3.1 Invoice Data Requirements

Field	Details
Invoice Type	Sale Invoice / Debit Note
Invoice Date	YYYY-MM-DD format
Seller Information	NTN/CNIC, Business Name, Address
Buyer Information	NTN/CNIC, Business Name, Address, Registration Type (Registered / Unregistered)
Items	HS Code, Description, Quantity, Rate, Tax, Discounts, SRO Schedule if applicable
Scenario ID	For Sandbox testing

3.2 Authentication / API Access

- Sandbox token for testing
- Production token for live submission
- IP whitelisting (minimum 1 IP from DPL infrastructure)

3.3 Operational Requirements

- User-friendly interface for Finance team
- Error logging and alerting
- Retry mechanism for failed submissions
- Audit trail for compliance

3.4 Infrastructure Options

- **Simple:** Postman / API client with manual JSON editing
- **Intermediate:** Streamlit / desktop app with token & JSON generation
- **Full AWS Integration:** Lambda + API Gateway + DynamoDB + S3 + CloudWatch

4. Integration Steps (Proposed)

Phase 1 – Requirements Gathering

- Identify exact invoice fields needed by Finance
- Confirm ERP / QuickBooks version
- Determine invoice volume and workflow
- Define user access & roles

Phase 2 – Sandbox Setup

- Register DPL system in FBR IRIS portal
- Provide technical contact info & IP whitelisting
- Receive sandbox token
- Test submitting sample invoices (scenario-based)

Phase 3 – Tool / App Development

- **Postman Option:** Create template JSONs for Finance team
- **App Option:**
 - Simple GUI for data entry
 - Automatic JSON generation
 - Submit invoices via FBR API
 - Display FBR validation response

Phase 4 – Production Go-Live

- Request production token from FBR

- Confirm sandbox testing is successful
- Enable invoice submission in production
- Maintain logs and audit trail

Phase 5 – Monitoring & Support

- Monitor submission errors and retries
- Maintain a submission dashboard for Finance
- Update JSON templates if FBR changes API

5. Next Actions / Clarifications Needed

- Confirm mandatory fields for DPL invoices
- Determine monthly / daily invoice submission volume
- Identify owner for token management and error monitoring
- Confirm whether every submission needs to be logged for audit compliance